



Figure 4: End-to-end architecture for MongoDB (Image source: googleimages.com)

- It provides the server as a separate program for a client/server networked environment.

Cons

- Different members are unable to fix bugs and craft patches.
- Users feel that MySQL no longer falls under the category of a free OS.
- It's no longer community driven.
- It lags behind others due to its slow updates.

SQLite

SQLite is supposedly one of the most widely deployed databases in the world. It was developed in 2000 and, since then, it has been used by companies like Facebook, Apple, Microsoft and Google. Each of its releases is carefully tested in order to ensure reliability. Even if there are any bugs, the developers of SQLite are quite honest about the potential shortcomings by providing bug lists and the chronologies of different code changes for every release.

Pros

- It has no separate server process.
- The file format used is cross-platform.
- It has a compact library, which runs faster even with more memory.
- All its transactions are ACID compliant.
- Professional support is also available for this database.

Cons

- It's not recommended for:
 - Different client/server applications.
 - All high-volume websites.
 - High concurrency.
 - Large datasets.

MongoDB

MongoDB was developed in 2007 and is well-known as the 'database for giant ideas.' It was developed by the people

behind ShopWiki, DoubleClick, and Gilt Group. MongoDB is also backed by a large group of popular investors such as The Goldman Sachs Group Inc., Fidelity Investments, and Intel Capital. Since its inception, MongoDB has been downloaded over 15 million times and is supported by more than 1,000 partners. All its partners are dedicated to keeping this free and open source solution's code and database simple and natural.

Pros

- It has an encrypted storage engine.
- It enables validation of documents.
- Common use cases are mobile apps, catalogues, etc.
- It has real-time apps with an in-memory storage engine (beta).
- It reduces the time between primary failure and recovery.

Cons

- It doesn't fit applications which need complex transactions.
- It's not a drop-in replacement for different legacy applications.
- It's a young solution—its software changes and evolves quickly.

MariaDB

MariaDB has been developed by the original developers of MySQL. It is widely used by tech giants like Facebook, Wikipedia and even Google. It's a database server that offers drop-in replacement functionality for MySQL. Security is one of the topmost concerns and priorities for MariaDB developers, and in each of its releases, the developers also merge in all of MySQL's security patches, even enhancing them if required.

Pros

- It has high scalability with easier integration.
- It provides real-time access to data.
- It has the maximum core functionalities of MySQL (MariaDB is an alternative for MySQL).
- It has alternate storage engines, patches and server optimisations.

Cons

- Password complexity plugin is missing.
- It does not support the Memcached interface.
- It has no optimiser trace. **END**

References

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